3 The influence of the zeros on the step or frequency responses

3.1 Step response

1. Plotting the step response. The MATLAB script assumes $G(s) = \frac{1}{(s+1)(0.5s+1)} = \frac{1}{s+1}\frac{1}{0.5s+1}$ and $G_c(s) = \frac{-s+c}{c(s+1)(0.5s+1)} = \frac{-s+c}{c}\frac{1}{s+1}\frac{1}{0.5s+1}$.

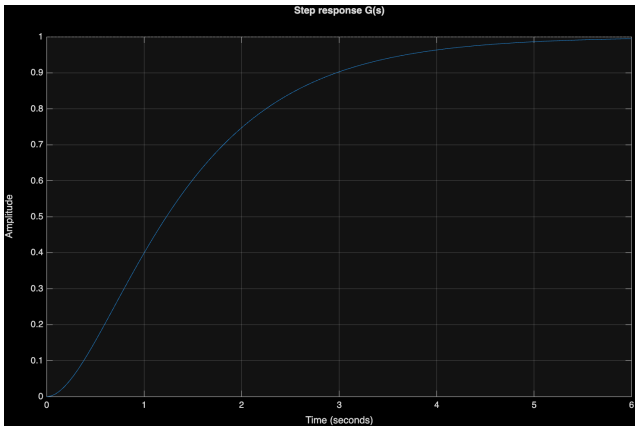


Figure 1: Unit-step response of $G(s)$

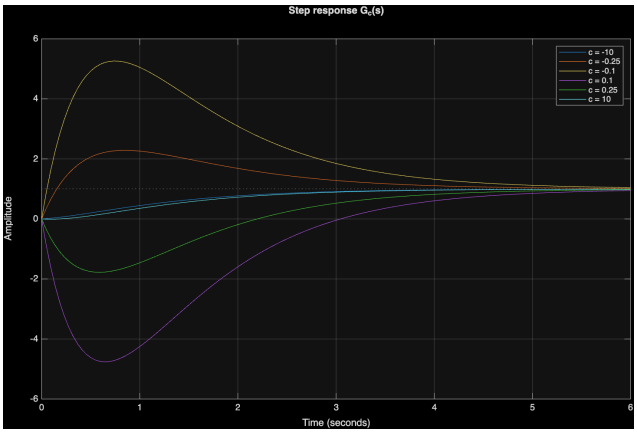


Figure 2: Unit-step responses for different zero locations c.

2. The overshoot and undershoot values are shown in the tables below.

Overshoot (%)	Undershoot (%)
0.00	0.00

Table 1: Overshoot and undershoot of $G(s)$.

c	Overshoot (%)	Undershoot (%)
-10.00	0.00	0.00
-0.25	128.53	0.00
-0.10	426.27	0.00
0.10	0.00	476.19
0.25	0.00	177.75
10.00	0.00	0.83

Table 2: Overshoot and undershoot of $G_c(s)$.

3.2 Frequency response